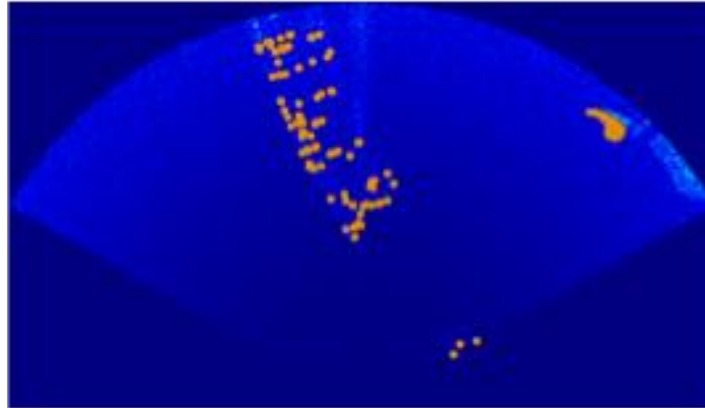
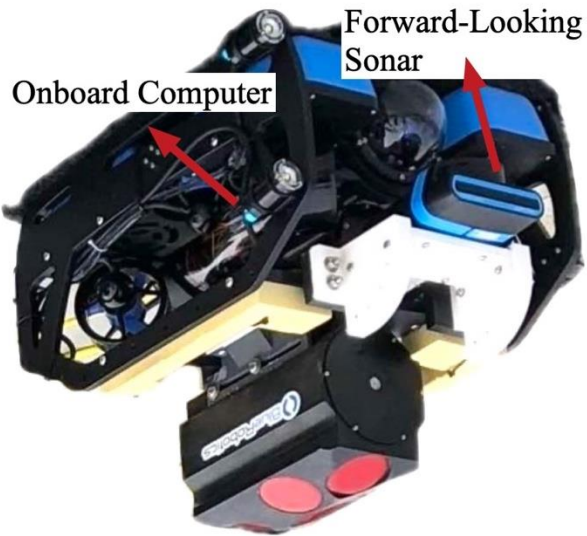


DRACo-SLAM2: Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams with Object Graph Matching

Yewei Huang*, John McConnell*, Xi Lin and Brendan Englot

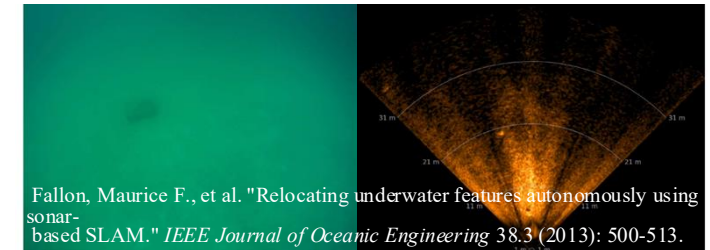


Main Challenges for Underwater Environments

- Even high-speed(hs) acoustic modems are limited to **62.5 kbps**.
- **Murky water** reduces visibility and can lead to failures in vision-based sensors.
- Long-term missions require systems that are **robust to mapping and localization failures**.



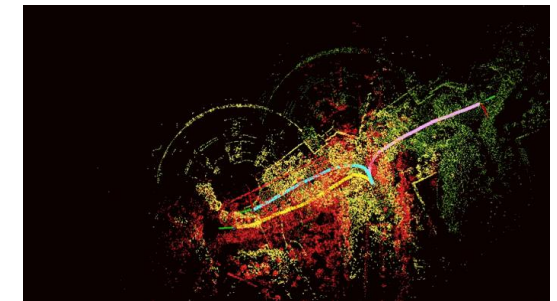
<https://www.evologics.com/acoustic-modems>
EvoLogics underwater acoustic modems.



Fallon, Maurice F., et al. "Relocating underwater features autonomously using sonar-based SLAM." *IEEE Journal of Oceanic Engineering* 38.3 (2013): 500-513.

The visual image (left) has limited visibility, while the sonar image (right) reveals three features.

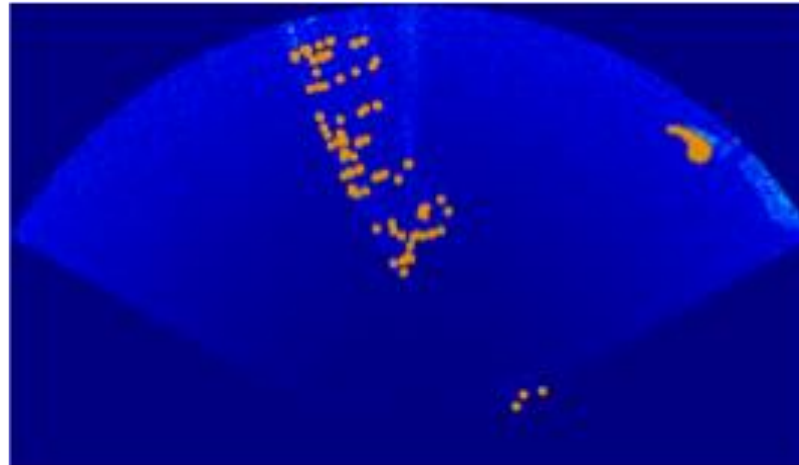
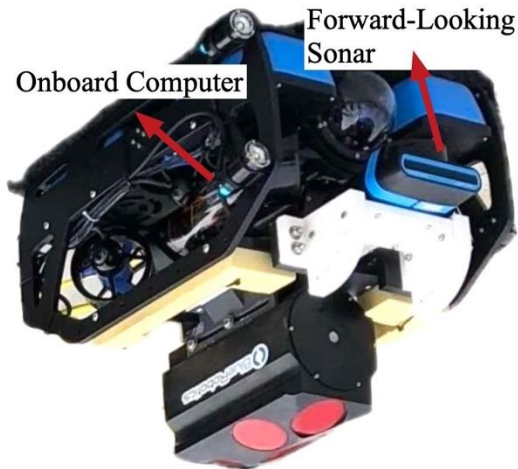
How to perform multi-robot SLAM
efficiently and **effectively** in such environments?



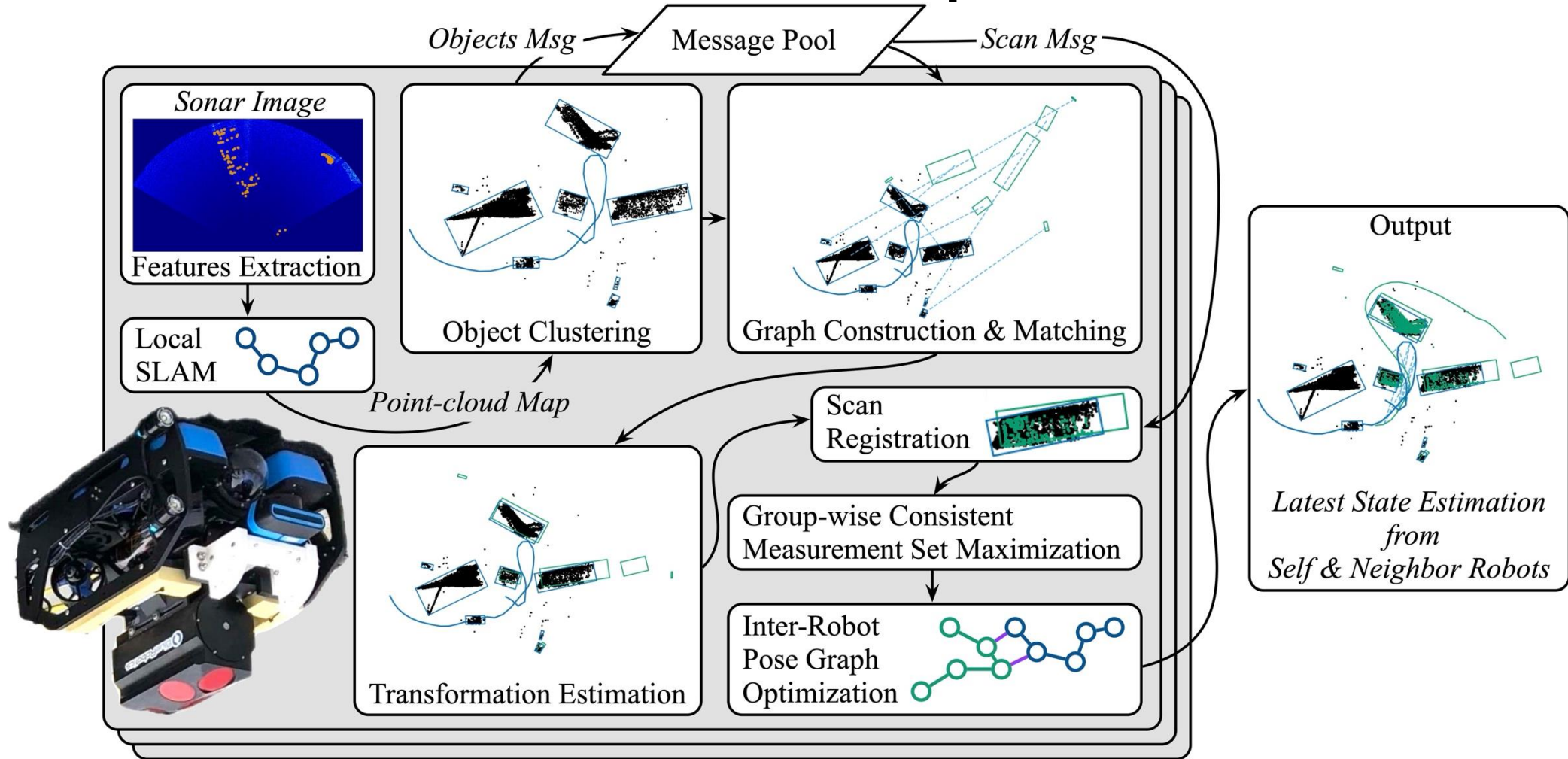
Map from different robots merged.

Our DRACo-SLAM2

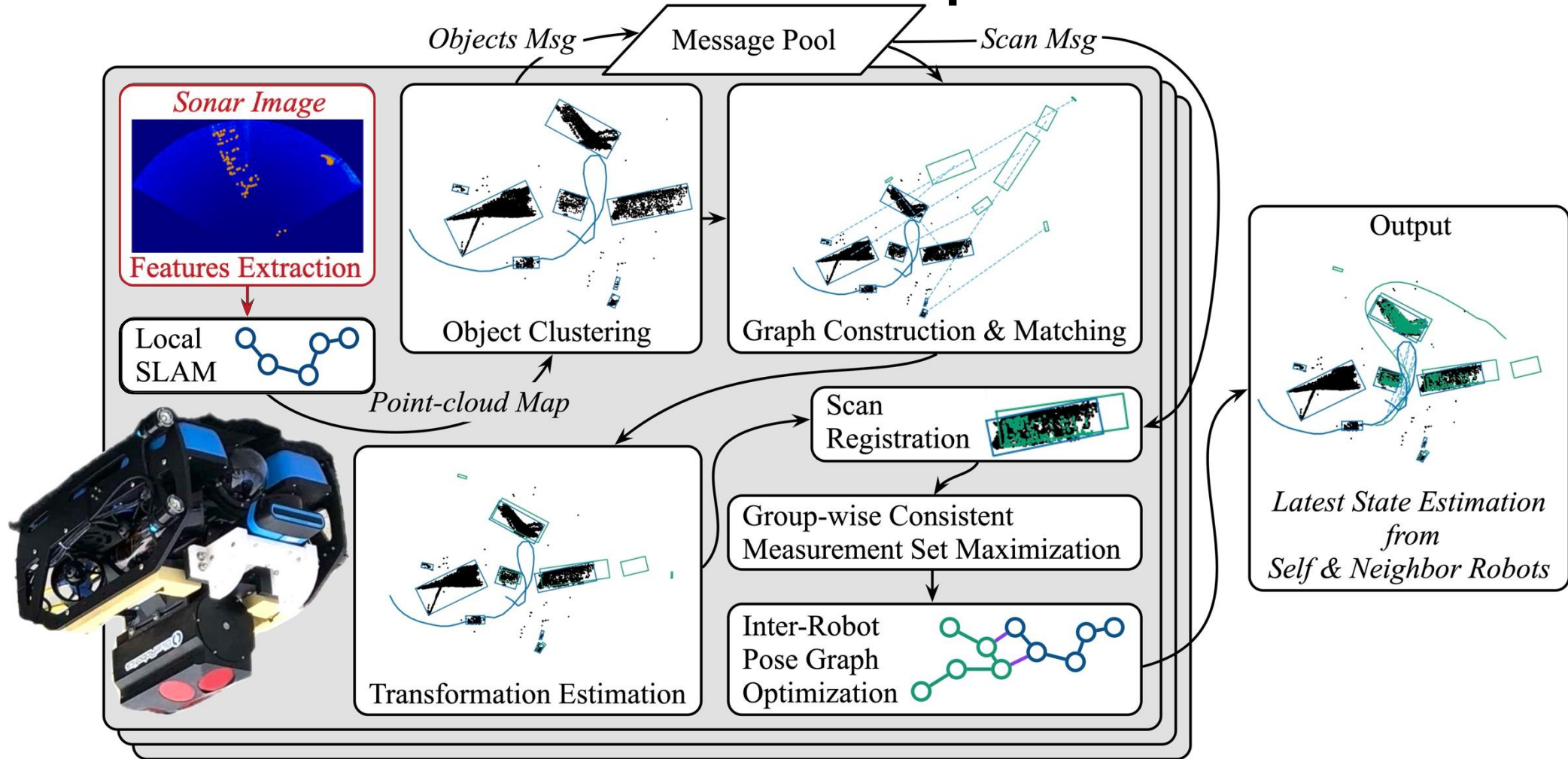
- **Object graph matching** for inter-robot loop closure detection using sonar images in underwater environments.
- A **reliable initial guess for ICP scan registration**, achieving a **5× speedup** over the previous version with Go-ICP.
- **Group-wise Consistent Measurement Set Maximization (GCM)**, providing robustness when multiple candidates exhibit similar registration errors.



An Overview of the DRACo-SLAM2 Pipeline

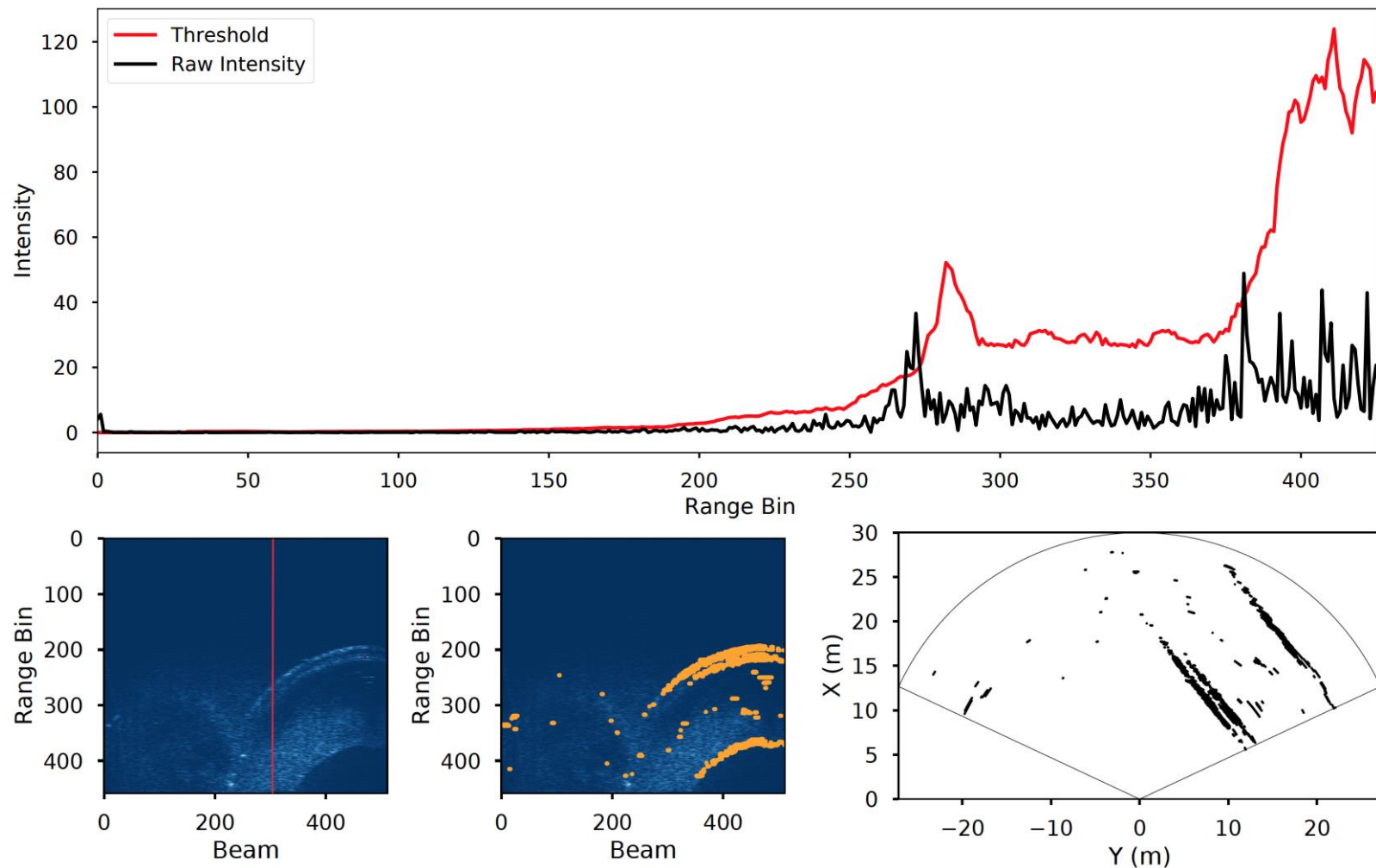


An Overview of the DRACo-SLAM2 Pipeline



Feature points are extracted from forward-looking sonar (FLS) images using **SOCA-CFAR**.

SOCA-CFAR (Smallest Of Constant False Alarm Rate)

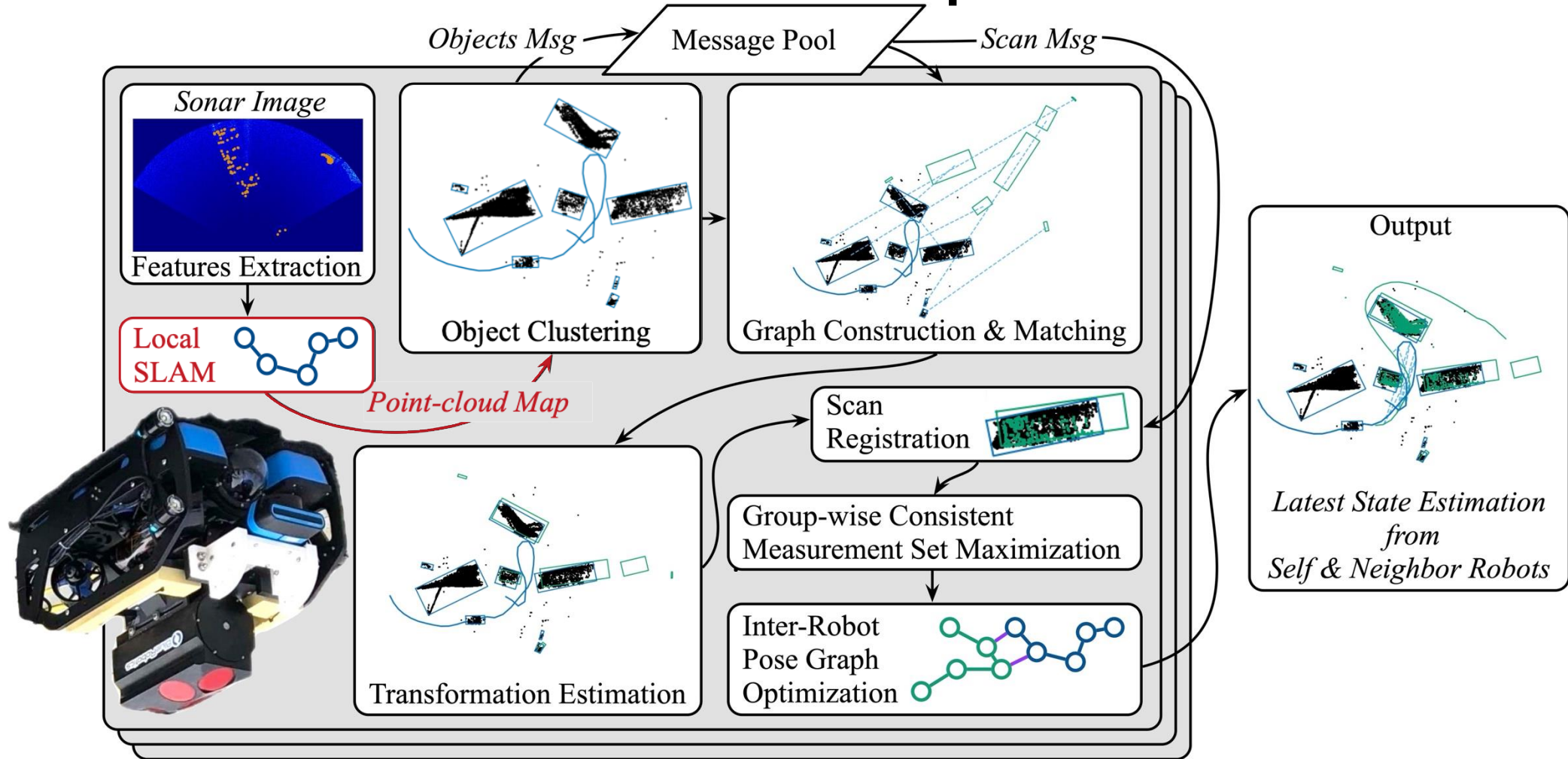


Feature Points in Polar Coordinate.

Feature Points in Cartesian Coordinate.

Sonar image from an Oculus M750d with 512 beams.

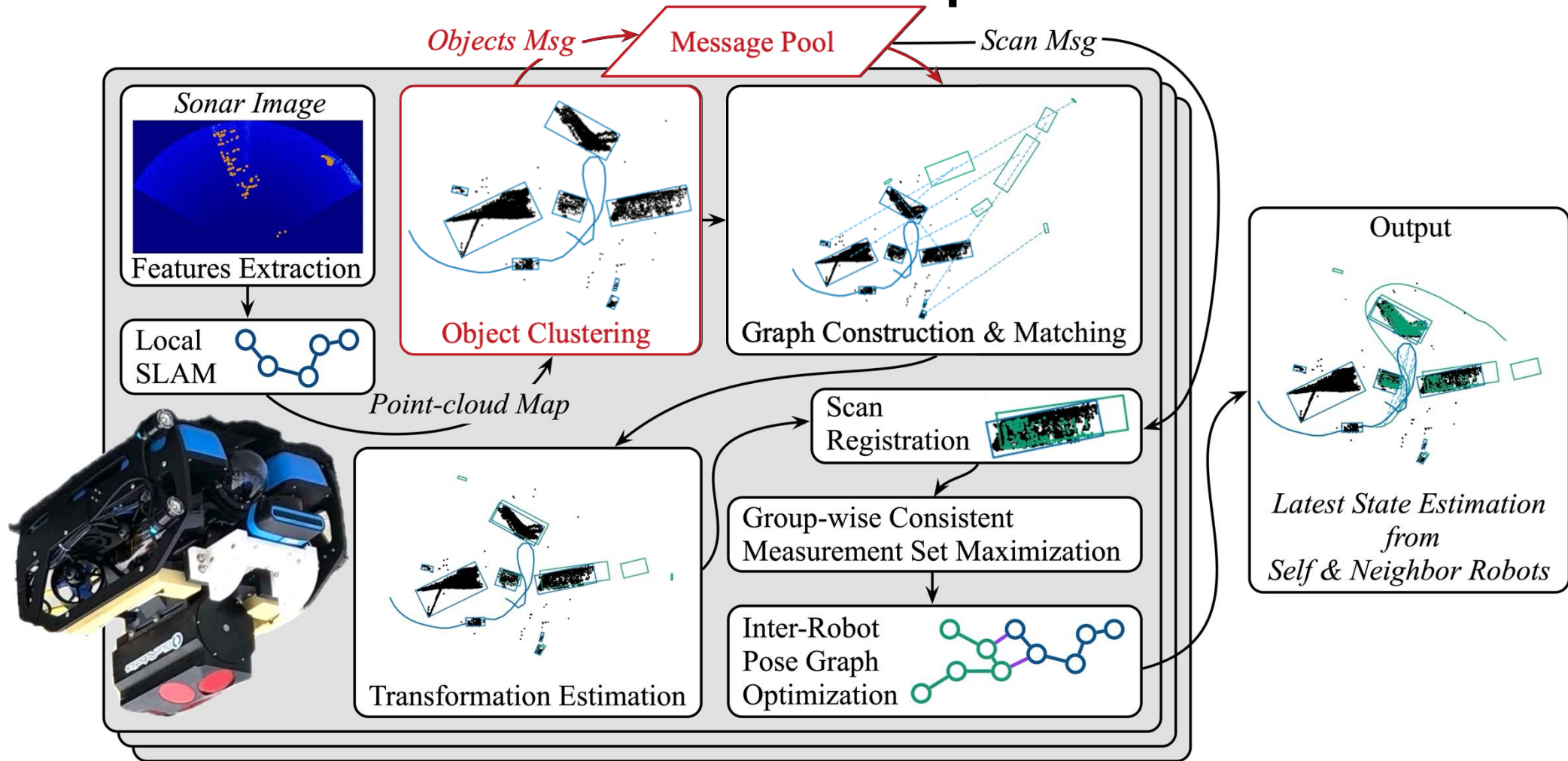
An Overview of the DRACo-SLAM2 Pipeline



Local SLAM (Bruce SLAM^[1])

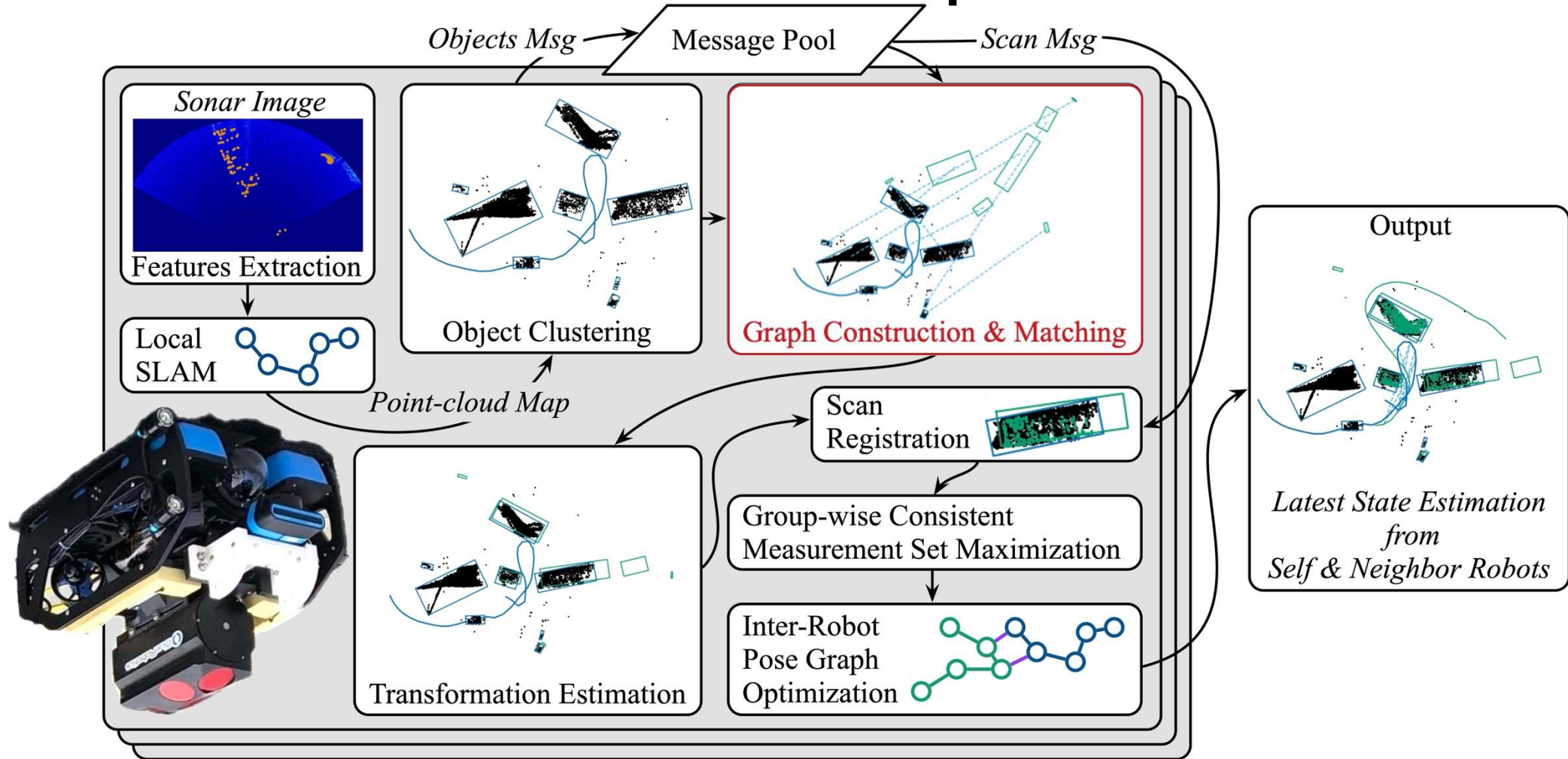
fuses sonar scan matching results with vehicle dead reckoning measurements.

An Overview of the DRACo-SLAM2 Pipeline



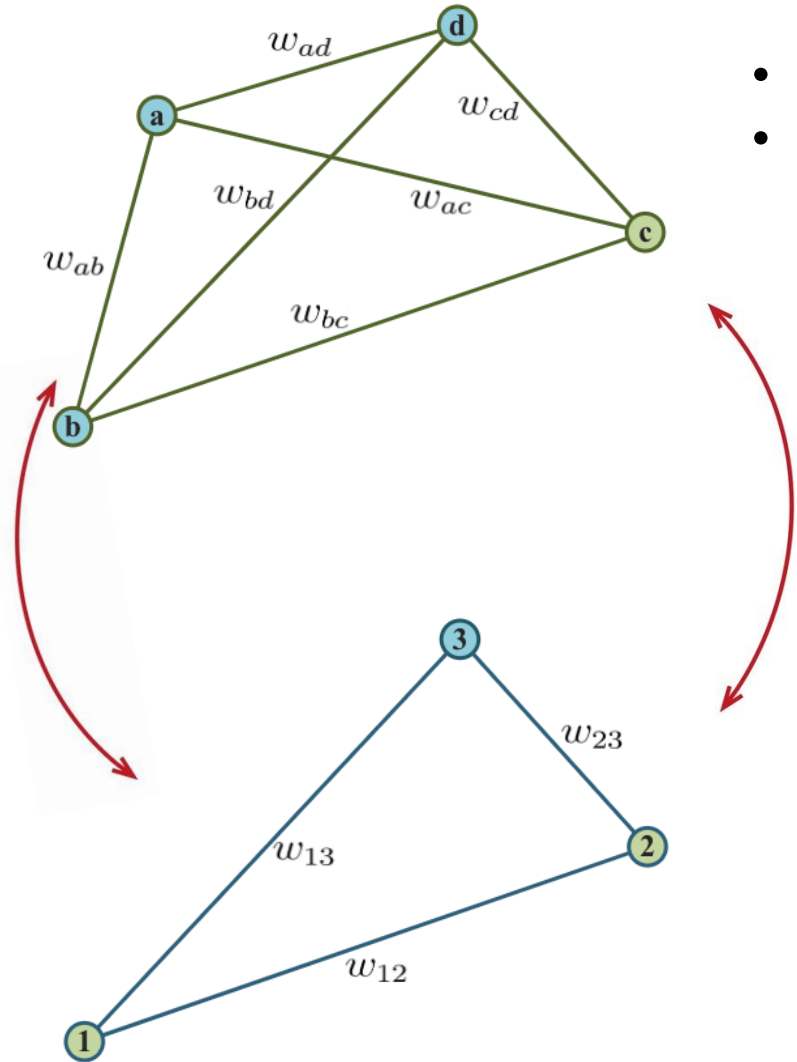
The local SLAM point-cloud map is clustered into an object map using **DBSCAN** and shared with neighbors.

An Overview of the DRACo-SLAM2 Pipeline

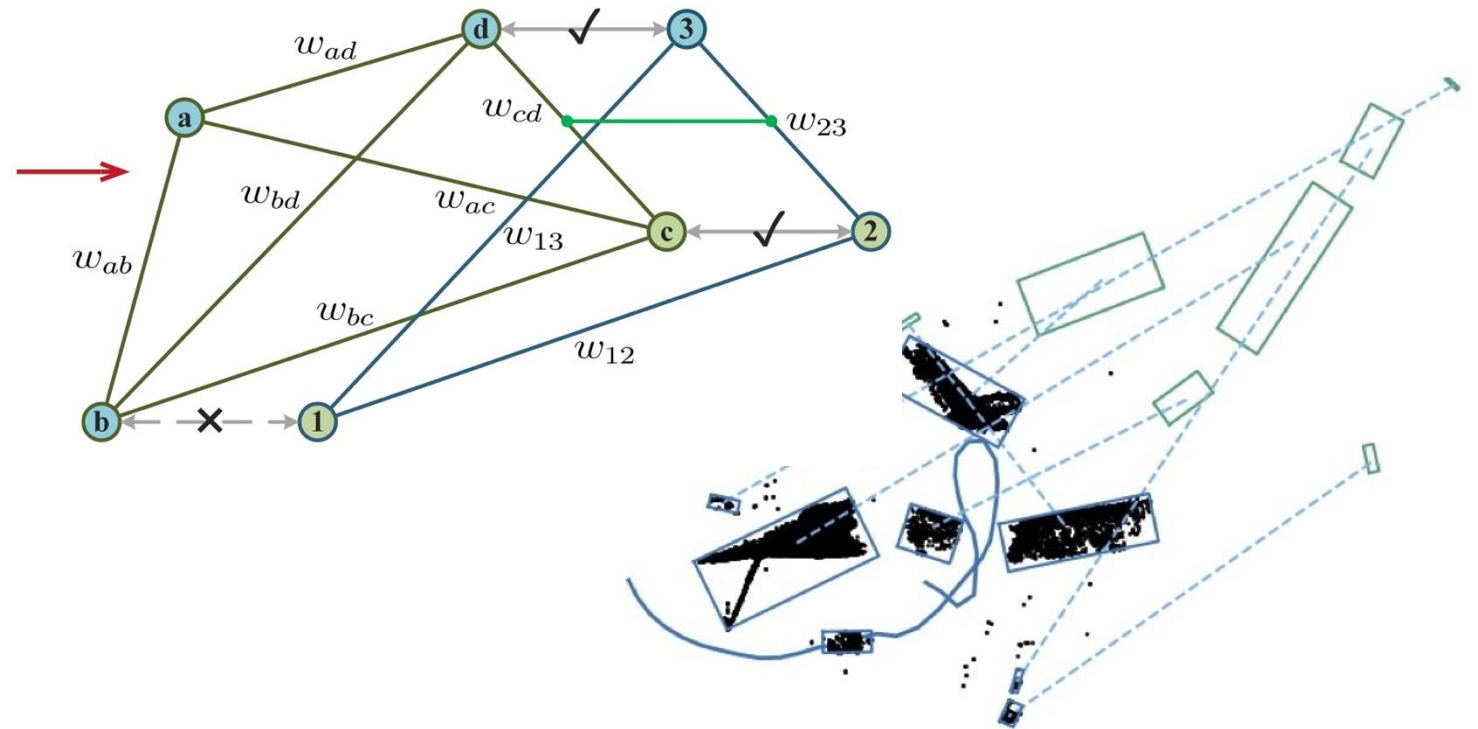


Fully connected object graphs are built from received object maps
and **matched** with the local object graph.

Inter-Robot Object Matching

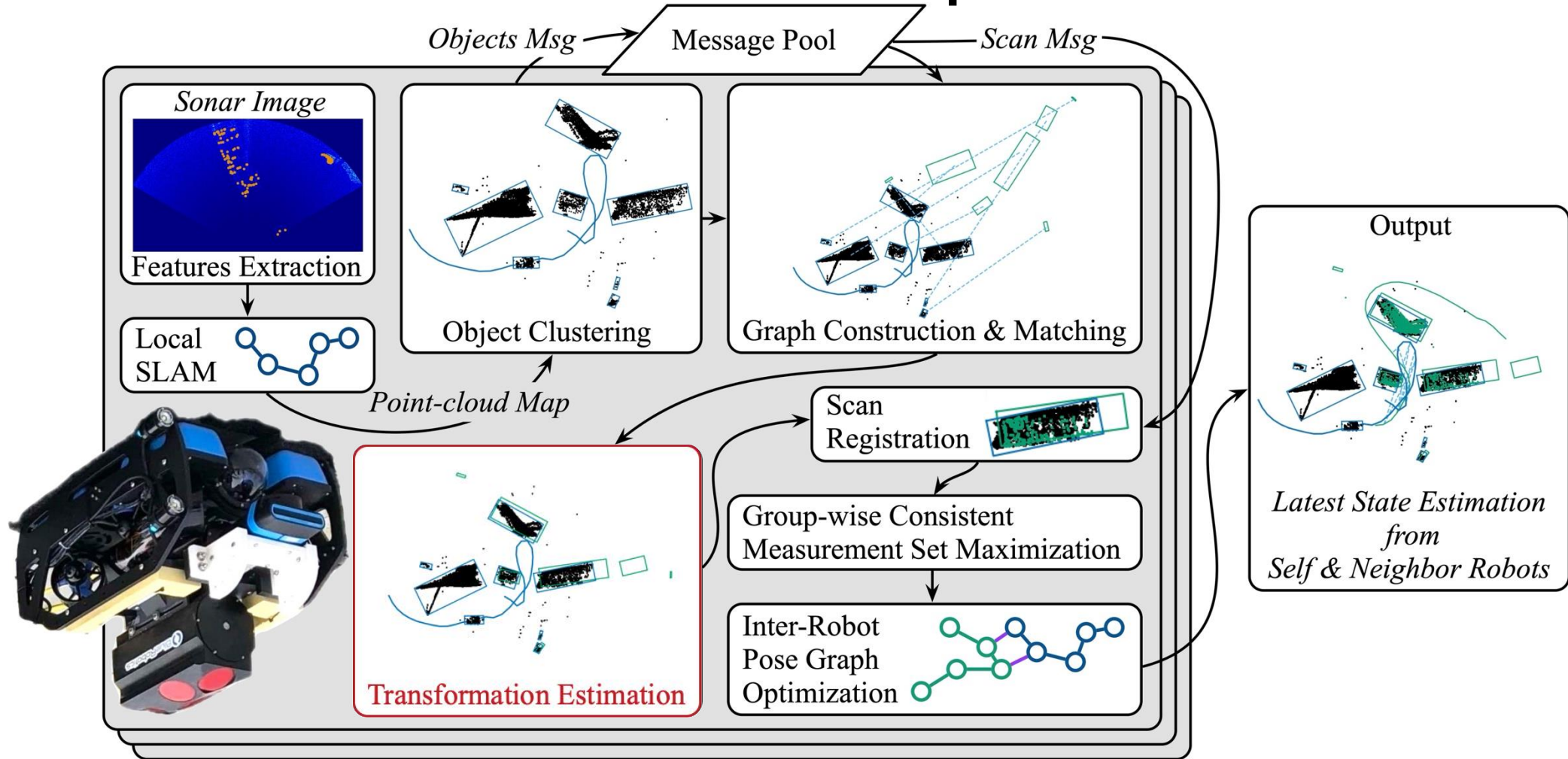


- Vertex-wise Comparison: Bounding box size;
- Edge-wise Comparison: Relative distance between object centers.



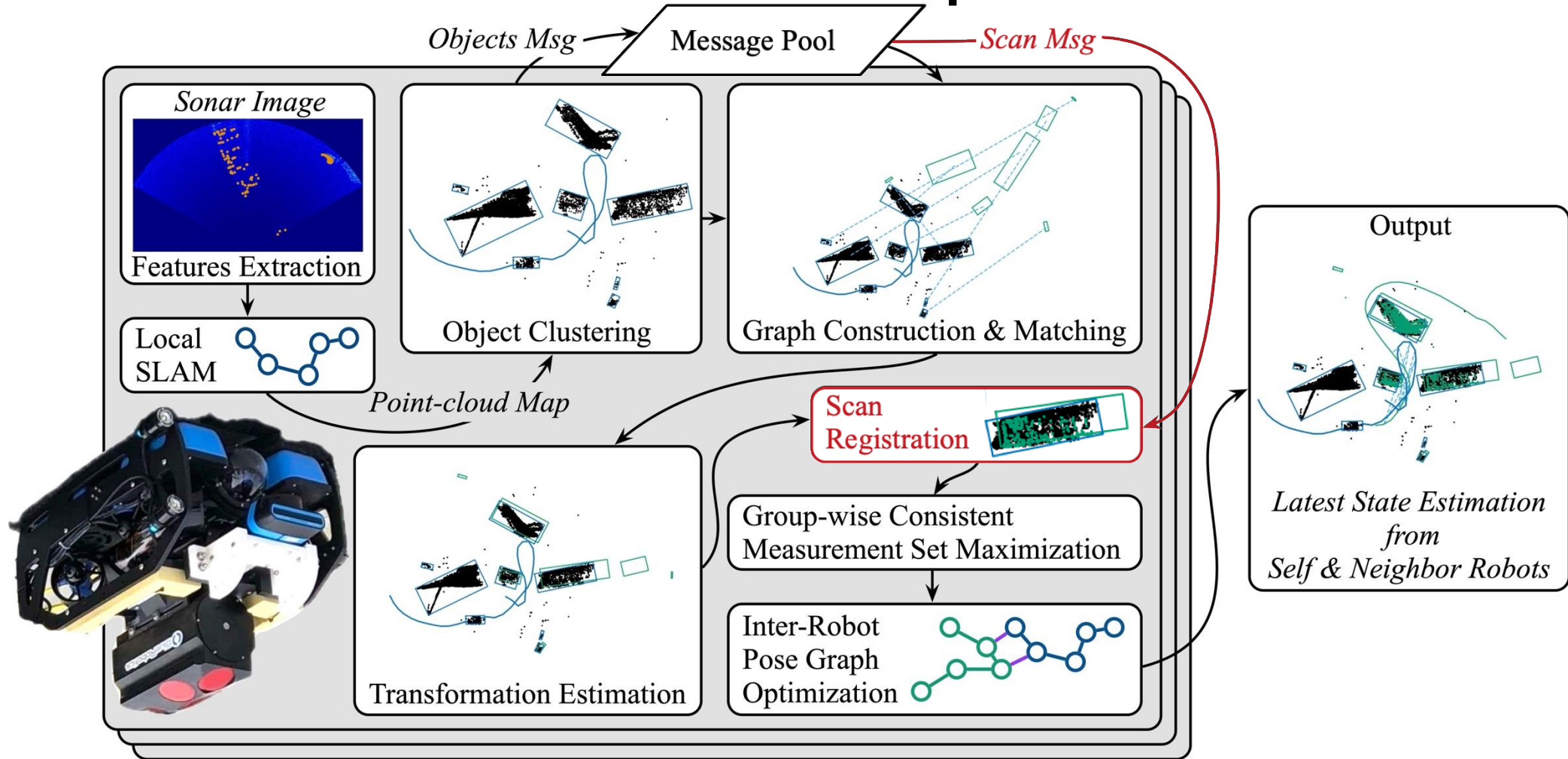
Matched results with local robot objects shown in blue and neighbor robot objects in green.

An Overview of the DRACo-SLAM2 Pipeline



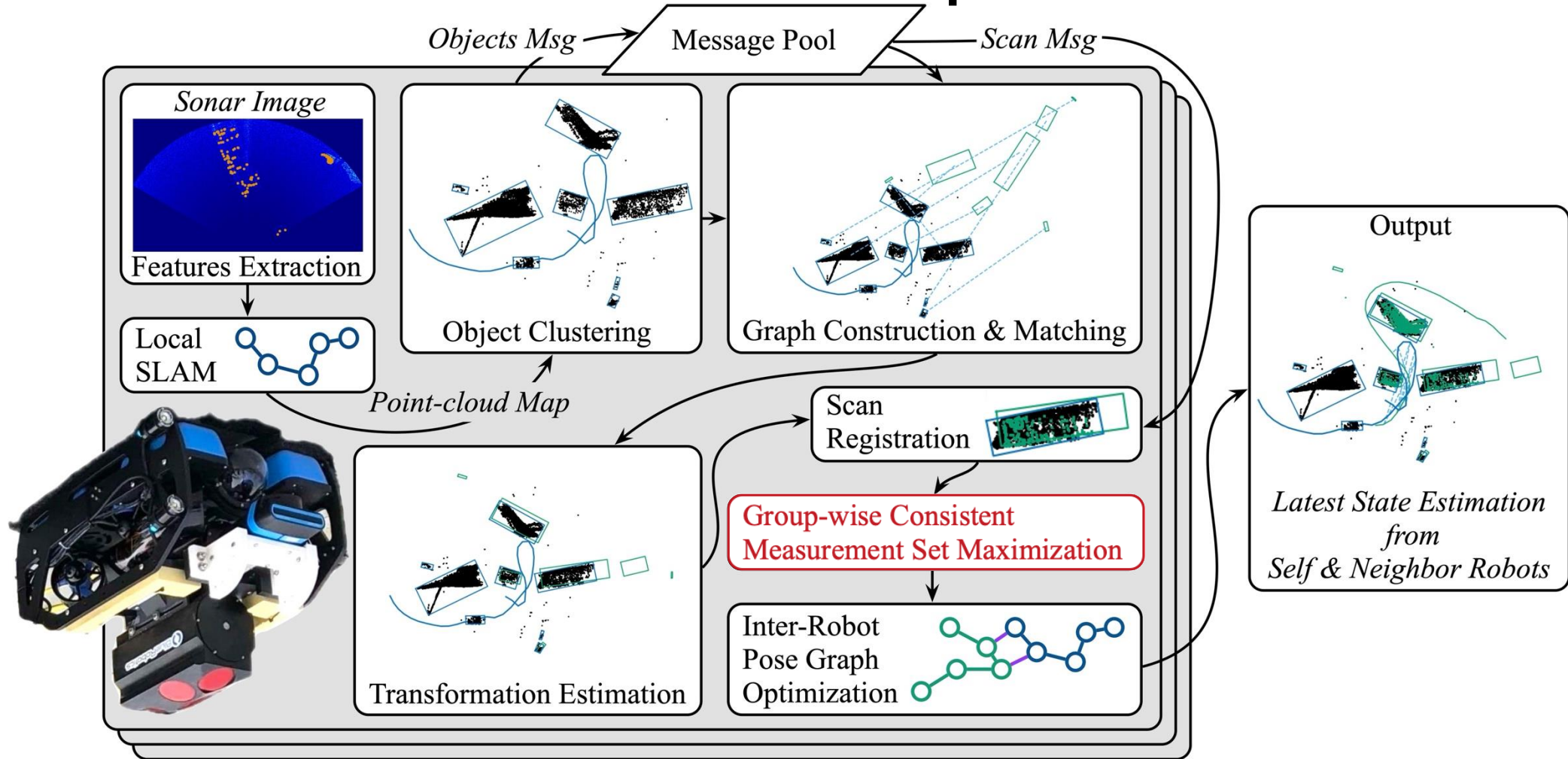
Once a neighbor graph matches the local graph,
the **transformation** between robot frames is estimated using **matched object centers**.

An Overview of the DRACo-SLAM2 Pipeline



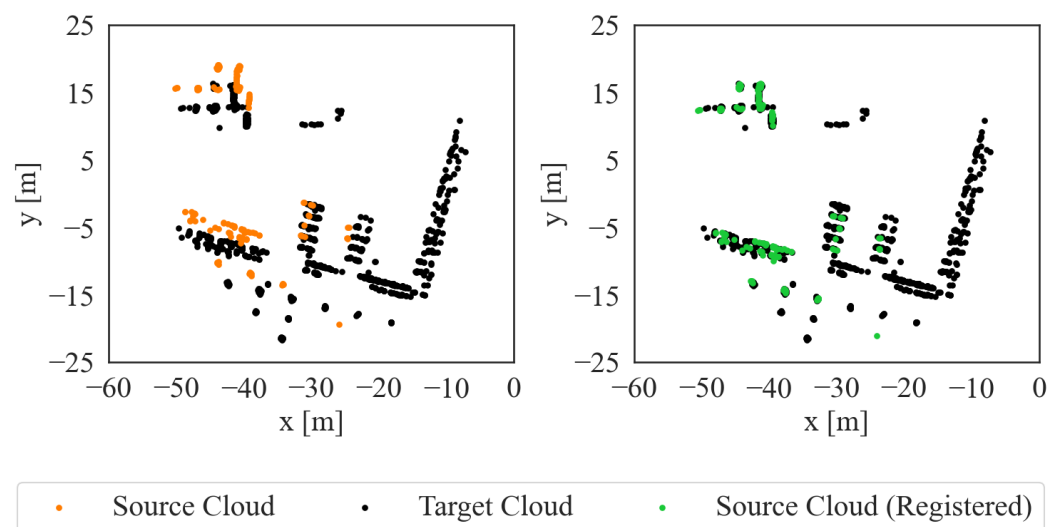
Sonar images are requested from neighbors,
and scan registration is performed using the estimated transformation as an initial guess.

An Overview of the DRACo-SLAM2 Pipeline

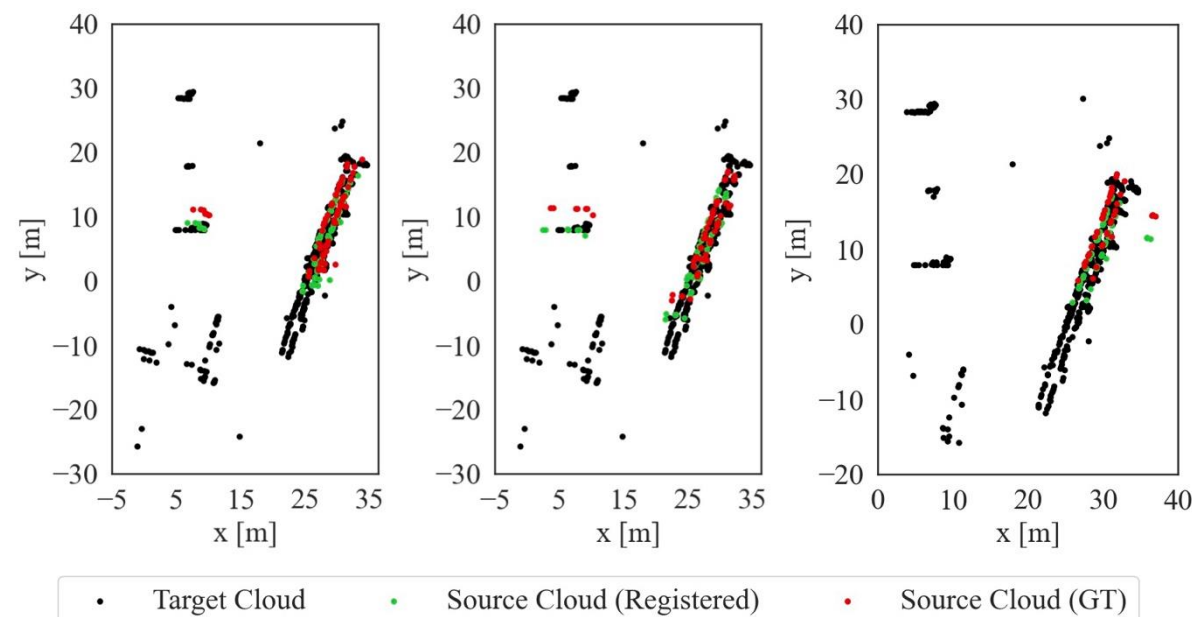


GCM is used to detect mis-associated loop closures.

Motivation: GCM



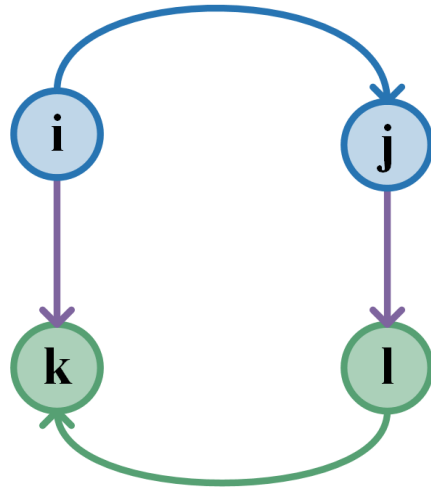
Success: The source cloud (orange) is roughly transformed into the local robot coordinate system using the transformation estimated from object graph matching.



Failure: A series of three inter-robot loop closures, in the same region of the environment are impacted by **similar point cloud registration errors**.

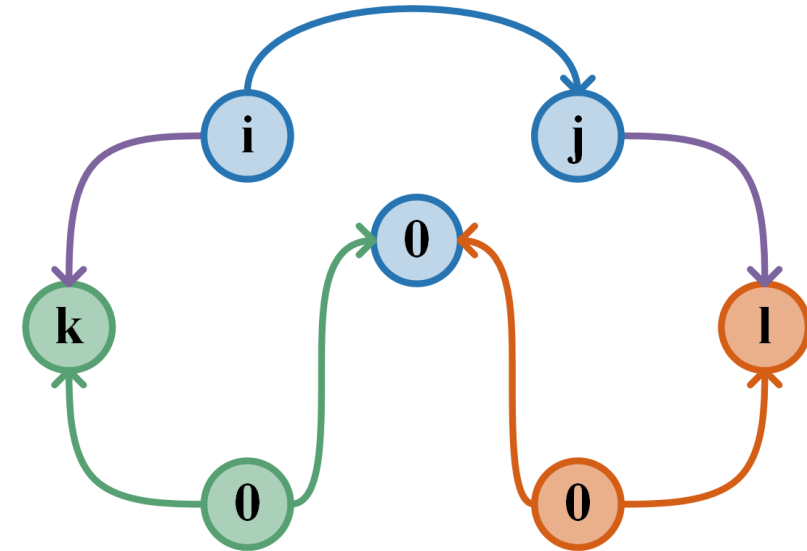
GCM is used to detect mis-associated loop closures.

GCM Algorithm



PCM

$$C(\mathbf{z}_{\alpha_k}^{\beta_i}, \mathbf{z}_{\alpha_l}^{\beta_j}) = \left\| \underbrace{(\mathbf{z}_{\alpha_k}^{\beta_i})^{-1}}_{\text{Inter-robot observation}} \cdot \underbrace{\hat{\mathbf{x}}_{\beta_j}^{\beta_i}}_{\text{Robot state estimation}} \cdot \mathbf{z}_{\alpha_l}^{\beta_j} \cdot \hat{\mathbf{x}}_{\alpha_k}^{\alpha_l} \right\|.$$



GCM

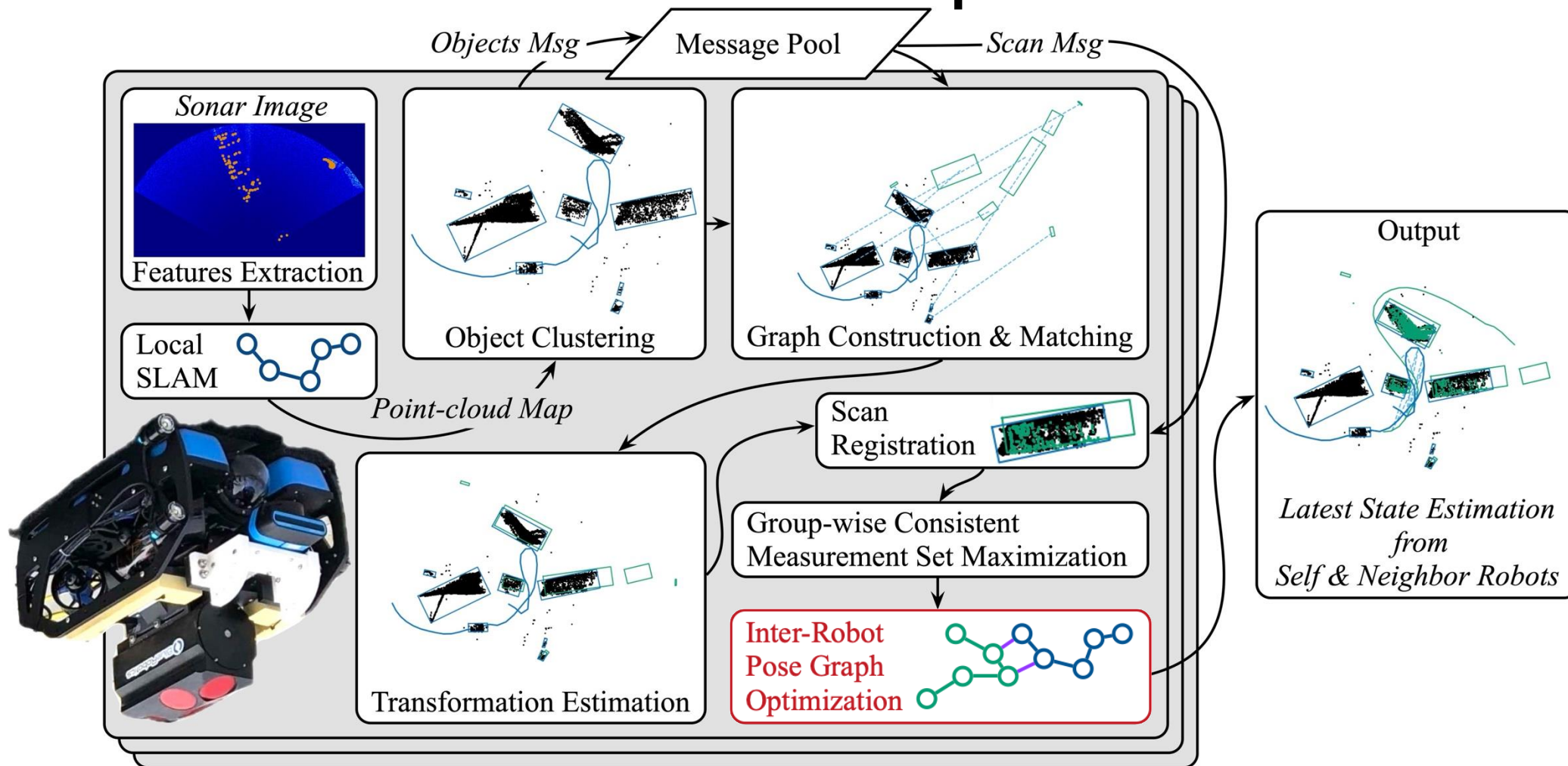
Robot coordinate
transformation
from scene graph
matching

$$\begin{aligned} \hat{\mathbf{x}}_{\gamma_j}^{\beta_i} &= (\hat{\mathbf{x}}_{\beta_i}^{\alpha_0})^{-1} \cdot \hat{\mathbf{x}}_{\gamma_l}^{\alpha_0}, \\ &= (\hat{\mathbf{x}}_{\beta_i}^{\beta_0})^{-1} \cdot \hat{\mathbf{x}}_{\alpha_0}^{\beta_0} \cdot \underbrace{(\hat{\mathbf{x}}_{\alpha_0}^{\gamma_0})^{-1}}_{\text{Robot coordinate transformation from scene graph matching}} \cdot \hat{\mathbf{x}}_{\gamma_j}^{\gamma_0}. \end{aligned}$$

$$C(\mathbf{z}_{\alpha_k}^{\beta_i}, \mathbf{z}_{\alpha_l}^{\gamma_j}) = \left\| (\mathbf{z}_{\alpha_k}^{\beta_i})^{-1} \cdot \hat{\mathbf{x}}_{\gamma_j}^{\beta_i} \cdot \mathbf{z}_{\alpha_l}^{\gamma_j} \cdot \hat{\mathbf{x}}_{\alpha_k}^{\alpha_l} \right\|.$$

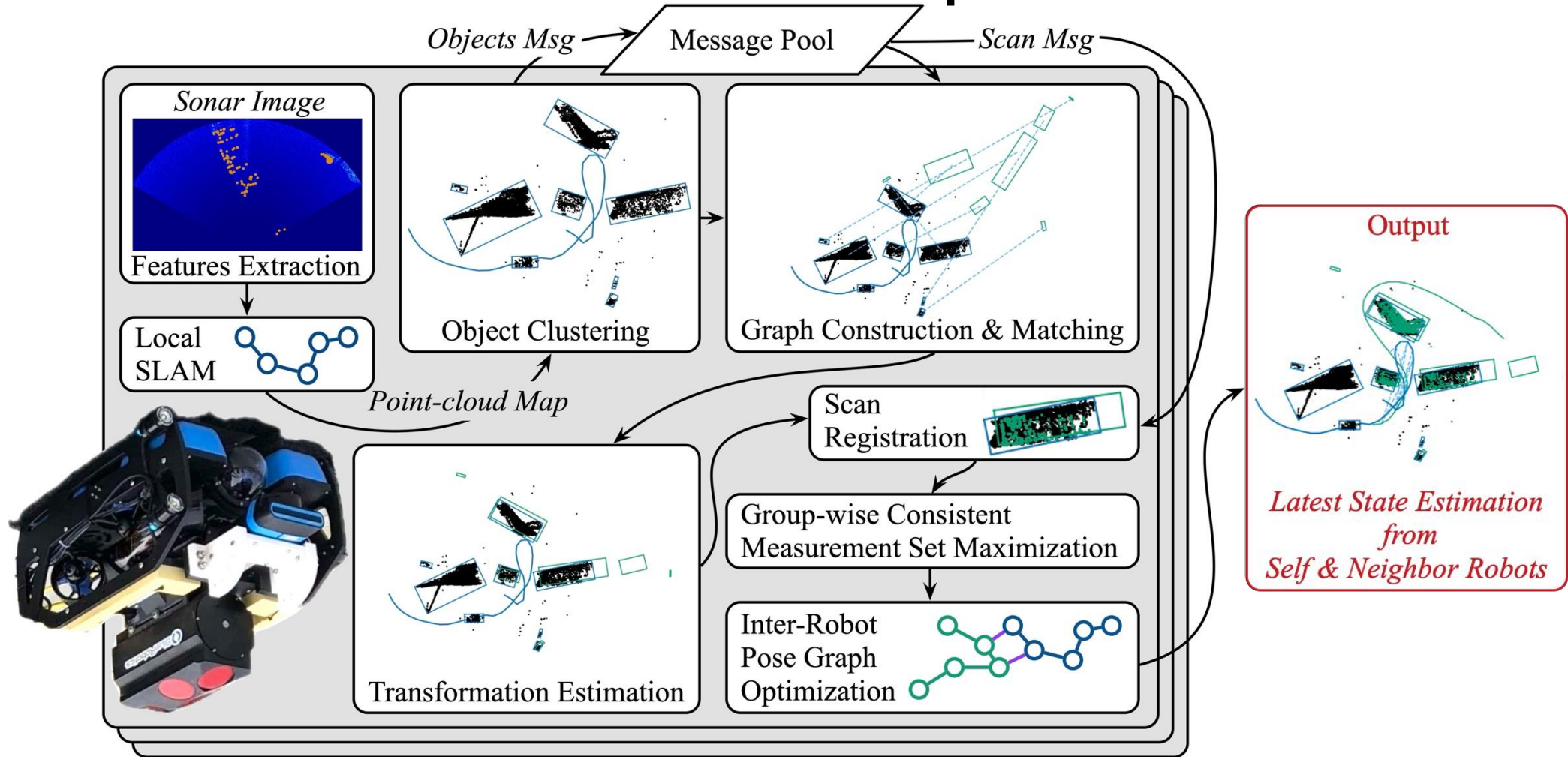
GCM is used to detect mis-associated loop closures.

An Overview of the DRACo-SLAM2 Pipeline



The loop closures accepted by GCM are added into the pose graph optimization.

An Overview of the DRACo-SLAM2 Pipeline



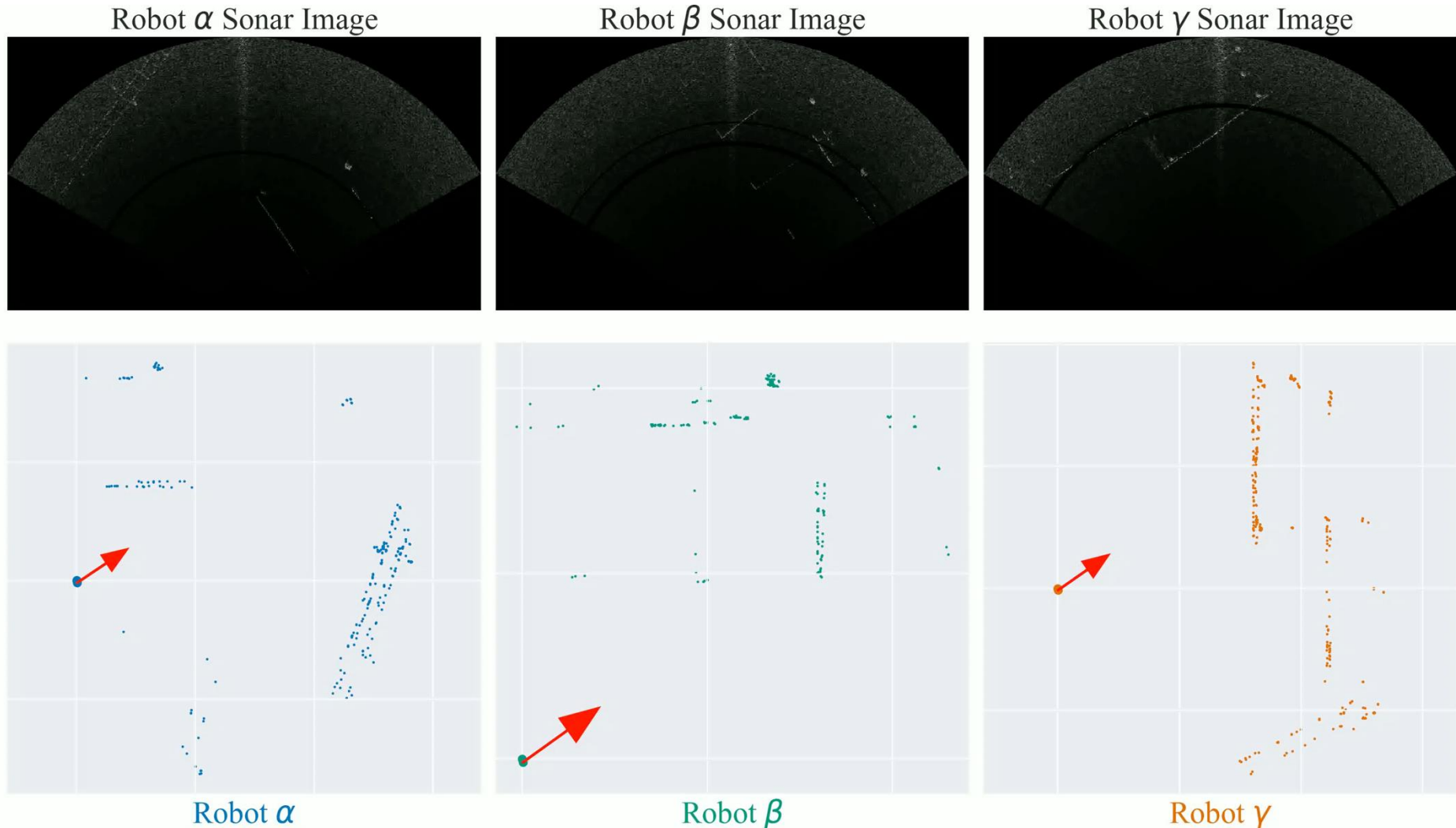
Finally, we obtain the refined state estimation of self and neighbor robots.

Experimental Setup

- Real-world Dataset:
 - Robot Type: customized BlueROV2-Heavy
 - Number of Robots: 3
 - Location: the U.S. Merchant Marine Academy (USMMA), King's Point, NY.
 - Sensors:
 - horizontally-oriented Oculus M750d sonar
 - Rowe SeaPilot DVL
 - VectorNav VN100 MEMS IMU.
- Simulation Datasets:
 - Environment: HoloOcean [\[1\]](https://frostlab.byu.edu/holocean-underwater-simulator)
 - Number of Robots: 3
 - USMMA Dataset
 - Airplane Dataset
- Experiment Computer:
 - Intel Xeon E-2276M CPU
 - 31.1 GB of memory
 - Ubuntu 20.04 with ROS Noetic
 - simulated communications between robots

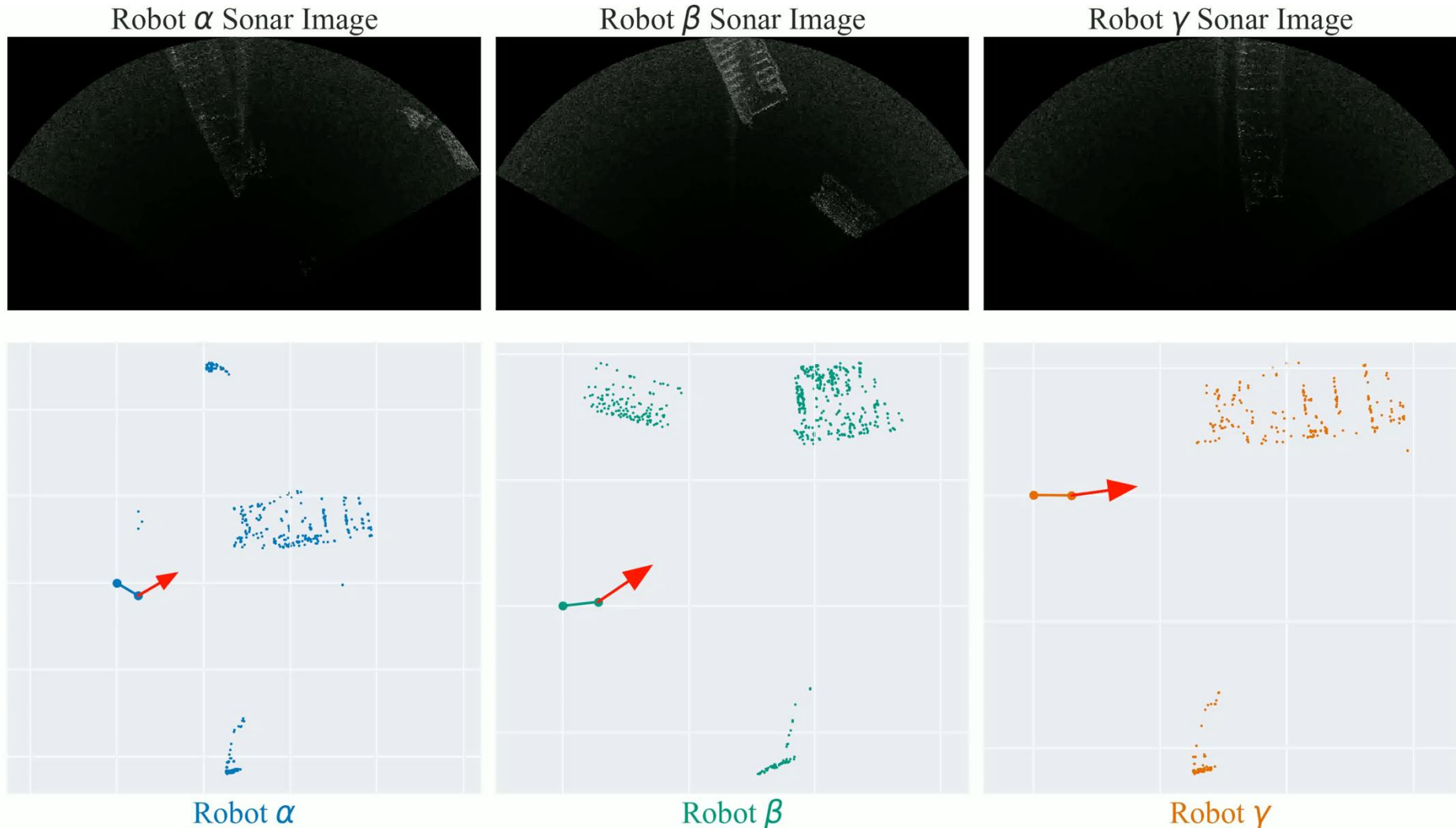
[1] <https://frostlab.byu.edu/holocean-underwater-simulator>

Performance on USMMA Simulation Dataset



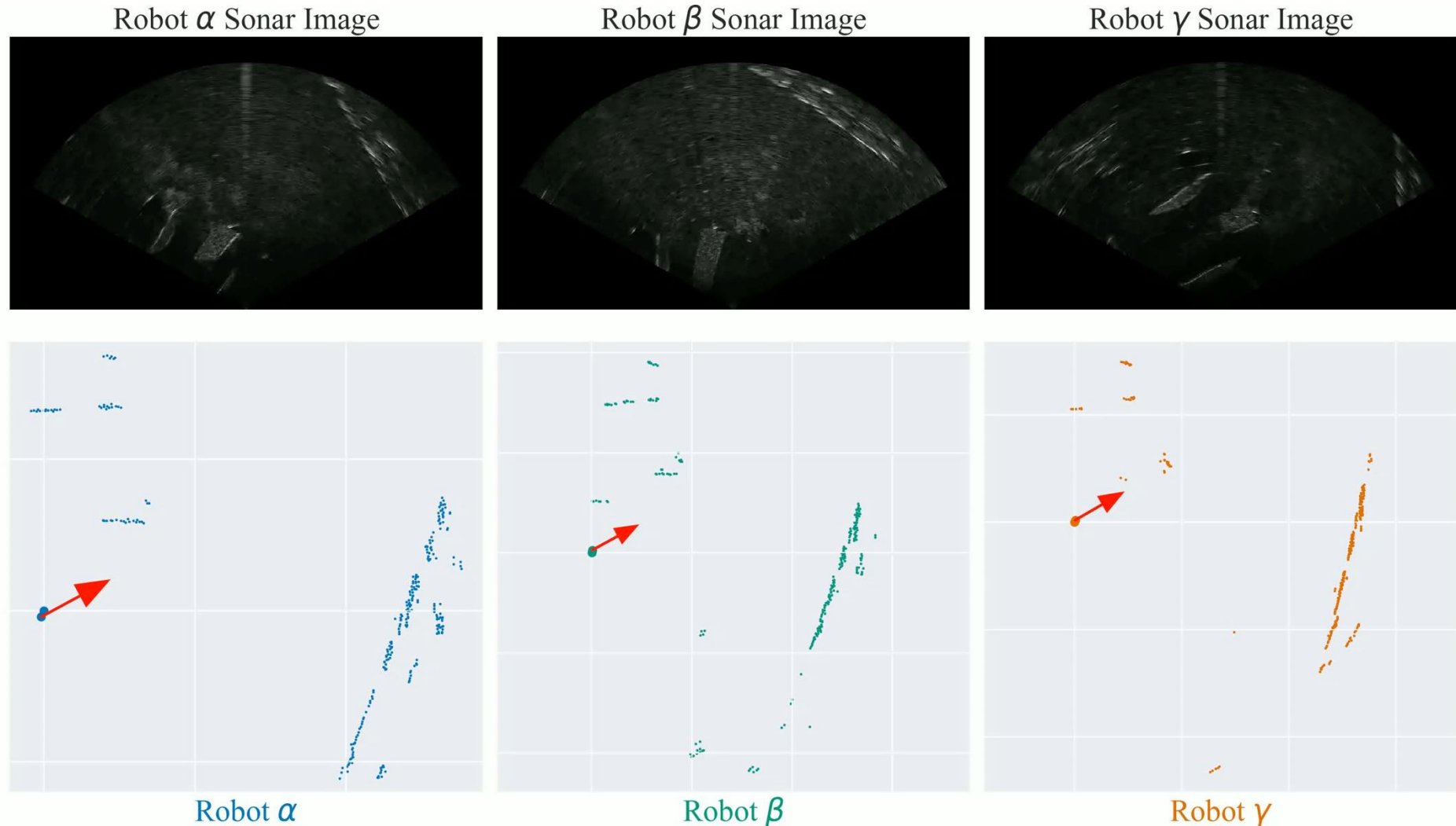
Shown above are all sonar image keyframes from this dataset, and the corresponding SLAM results.

Performance on Airplane Simulation Dataset



Shown above are all sonar image keyframes from this dataset, and the corresponding SLAM results.

Performance on USMMA Real Dataset



Shown above are all sonar image keyframes from this dataset, and the corresponding SLAM results.

Quantitative Analysis

- **5× faster** than the previous Go-ICP version.
- **Meter-level accuracy** comparable to the previous version.
- **10 Kbits mean / 28 Kbits max** message size, within **62.5 kbps** modem bandwidth.

Table 1: Mean runtime and mean number of inter-robot loop closure algorithm executions per time step for DRACo-SLAM (DRACo1) and DRACo-SLAM2 (DRACo2) on both of our 3-robot fully simulated datasets.

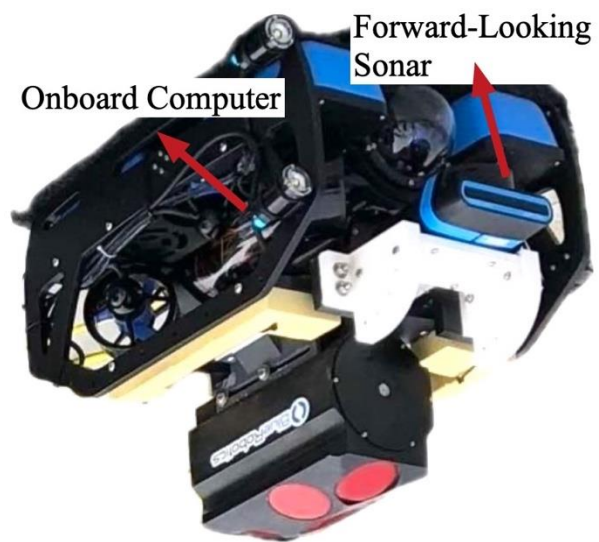
	Algorithm	Mean (s)	Mean No.
DRACo1	Ring Key Matching	< 0.01	1
	Go-ICP Registration	10.11	28
DRACo2	Graph Construction & Matching	< 0.01	1
	ICP Registration (Window Size 3)	1.55	122

Table 2: Absolute Trajectory Error (ATE) in meters for the proposed method compared to the full Pose Graph Optimization (PGO) method on both of our 3-robot fully simulated datasets.

Algorithm		USMMA			Airplane		
		α	β	γ	α	β	γ
DRACo	PCM	1.58	2.43	1.88	1.32	1.36	1.46
	GCM	1.43	1.20	1.23	1.33	0.95	1.31
Full PGO	PCM	1.27	1.37	1.52	1.28	1.91	1.63
	GCM	1.24	1.41	1.56	1.31	1.91	1.65

Table 3: Sizes of perception messages for DRACo1 and DRACo2 on the USMMA real-world dataset (with simulated 3-robot comms.).

Algorithm	Message Type	Mean KBits	Max KBits
DRACo1	Ring key Descriptor	0.13	0.13
	Point Cloud-float32	9.67	27.90
DRACo2	Object Map	1.35	2.25
	Point Cloud-float32	9.69	27.90



Thank you!

